Sean Luka Girgis

Big Data Platform Engineer | Hadoop, Spark, Linux Automation & Platform Reliability

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LinkedIn | GitHub | Portfolio

Professional Summary

Senior Data Engineer and platform-oriented technologist with 20+ years of enterprise experience across big data processing, Hadoop/Hive, Spark/PySpark, Linux/Unix automation, telemetry platforms, performance engineering, monitoring, and production reliability. Experienced acting as a bridge between operations, engineering, infrastructure, and application teams to improve platform stability, resource utilization, observability, documentation, and operational readiness. Strong fit for Hadoop/Spark platform strategy roles requiring technical leadership, cluster health awareness, capacity optimization, scripting automation, and production support discipline.

Professional Experience

Senior Capacity & Data Engineer at CITI (2017-11 – 2025-12)

- Built and supported large-scale telemetry data pipelines using Python, Pandas, PySpark, AWS Glue, S3, Athena, Redshift, and Oracle to process infrastructure metrics from 6,000+ endpoints.
- Applied Hadoop/Hive, Spark, and distributed data processing concepts to large telemetry and forecasting workloads, improving scalable transformation and analytics readiness.
- Served as a liaison across infrastructure, application, monitoring, and stakeholder teams to translate operational needs into reliable data workflows, reporting layers, and platform improvements.
- Monitored production data pipelines, investigated failures, reconciled source feeds, and improved reliability across BMC TrueSight, CA Wily, AppDynamics, and Oracle reporting environments.
- Optimized resource usage and capacity planning through data mining, utilization analysis, forecasting, and infrastructure trend reporting across enterprise banking systems.
- Designed historical retention and schema strategies for large operational datasets, supporting seasonal forecasting, reporting, and performance analysis.
- Created documentation, runbooks, support patterns, and operational reporting to improve maintainability and reduce manual handoffs.
- Used Python, SQL, Linux/Unix scripting, and automation patterns to reduce repetitive reporting work and support production data operations.

Performance Engineer at G6 Hospitality LLC (2017-03 – 2017-11)

- Managed Dynatrace AppMon and Gomez Synthetic Monitoring platforms for Brand.com and critical hospitality systems, supporting platform health, transaction visibility, and production troubleshooting.
- Led the FAST project, mining real-user and synthetic performance data to identify optimization opportunities and communicate findings to engineering stakeholders.
- Supported monitoring platform upgrades and TLS 1.2 security enforcement with zero production monitoring regression during cutover.
- Built operational dashboards and reporting outputs that improved visibility into performance trends, service behavior, and reliability risks.

SME for CA APM (Senior Consultant) at CA Technologies / TIAA-CREF (2011 – 2016)

- Served as enterprise APM SME managing 50+ Enterprise Managers and 4,000–6,000 instrumented agents, supporting large-scale telemetry operations and production visibility.
- Led major platform upgrades from CA APM 9.1 to 10.1 with zero unplanned downtime, coordinating technical impacts across cross-functional IT teams.
- Built Perl and Korn shell automation for telemetry extraction and reporting, replacing manual exports with repeatable operational workflows.
- Defined dashboards, alerts, SLA thresholds, documentation, and operational practices adopted as standards across client environments.
- Collaborated with J2EE/WebLogic, infrastructure, and operations teams to troubleshoot complex platform and application performance issues.

Performance Test Engineer at AT&T (2010-08 – 2011-07)

- Analyzed large J2EE telecom systems under load to identify throughput limits, JDBC bottlenecks, memory pressure, CPU behavior, and thread contention.
- Installed and configured JMX Monitoring, Wily Introscope, and thread-dump automation to improve runtime visibility during performance testing.
- Documented performance baselines, operational thresholds, and regression findings used by engineering teams for release decisions.

Senior Systems & Data Migration Engineer at Sabre (2008-05 – 2010)

- Led migration of a high-throughput shopping engine from 200+ MySQL nodes to a 6-node Oracle RAC cluster while maintaining data integrity and transaction performance.
- Optimized C++/OCCI transaction processing and Oracle access patterns, reducing hardware footprint by 95% while supporting high-volume workloads.
- Built validation and regression testing utilities to verify migration correctness across millions of transaction records.
- Automated schema conversion, data validation, and phased cutover activities for a mission-critical enterprise platform migration.

Developer / Support Engineer (Billing, Interfaces, CSM/PRMS) at Corpus Inc. / Sprint (2001 – 2007)

- Developed and supported high-availability C++/Pro*C/PL/SQL billing interfaces for telecom-scale transaction systems.
- Improved database performance by 10x through SQL, sequence, and process optimization while reducing memory usage by 75%.
- Built Unix/Linux automation using Korn shell, Perl, Awk, Syncsort, and SQL to support reporting, deployments, housekeeping, and operational data processing.
- Supported production issue triage, QA cycles, release support, documentation, and cross-team troubleshooting across billing and customer-management systems.

Education

Post-Graduate Diploma in Computer Engineering Technology / Computer Science
Humber College, Toronto, Canada

Bachelor of Science in Civil Engineering
Zagazig University, Egypt

Skills

Flagship Projects

HorizonScale — AI Capacity Forecasting Engine
Built a capacity forecasting and telemetry analytics workflow using Python, PySpark, Prophet, scikit-learn, and Streamlit to process infrastructure metrics, identify utilization trends, and support proactive capacity planning.
Technologies: Python, PySpark, Spark, Prophet, scikit-learn, Streamlit

Serverless Lakehouse Platform (AWS)
Designed a cloud data platform pattern using S3, Glue, Athena, Redshift, PySpark, and Parquet for raw ingestion, transformation, curated analytics datasets, and scalable querying.
Technologies: AWS S3, AWS Glue, Athena, Redshift, PySpark, Parquet

AI-Powered Job Search Pipeline
Built a Python-based automation pipeline with semantic duplicate detection, LLM-based job scoring, resume/cover generation, quality gates, and status tracking using FAISS, sentence-transformers, and LLM APIs.
Technologies: Python, FAISS, Sentence Transformers, LLM APIs, Automation, JSON